Practice workbook · Explain how food moves by peristalsis and why fibre helps

Movement of the bolus along oesophagus and gut by muscle waves, and the role of dietary fibre.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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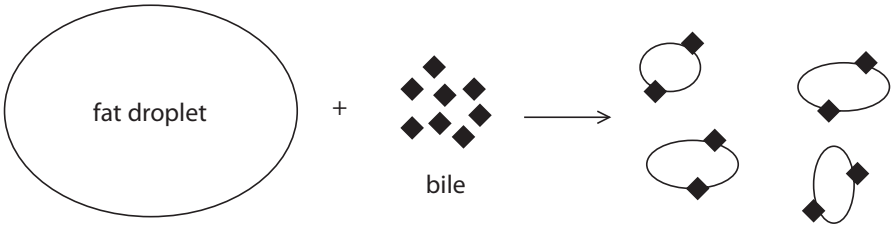
Type ID: HBI-PERISTALSIS-FIBRE

1. Question 3 · part c · 5 marks

January 2023 Paper 1

Worked-example retry · Attempt part c only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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1. Question 3 · part c · 5 marks

January 2023 Paper 1

Worked-example retry · Attempt part c only.

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(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

Reagent used	Colour of the food solution after the test
iodine solution	blue/black
biuret solution	purple
Benedict's solution	blue

(i) Describe what the food tests show about the dietary components in the food.

(3)

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(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

(3)

1. Question 3 · part c · 5 marks

January 2023 Paper 1

Worked-example retry · Attempt part c only.

(c) Food is digested as it moves along the alimentary canal.

This process is helped by a diet containing fibre.

Explain how food is moved along the alimentary canal and how fibre helps this process.

(5)

(Total for Question 3 = 15 marks)

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1. Question 3 · part c · 5 marks · Working and answer

Working and answer

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2. Question 1 · part c · 3 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

1 Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

(3)

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

(1)

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(ii) Describe the processes that occur in the mouth to digest food.

(4)

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2. Question 1 · part c · 3 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

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(c) Fibre in food cannot be digested.

Explain the importance of fibre in the diet.

(3)

(Total for Question 1 = 11 marks)

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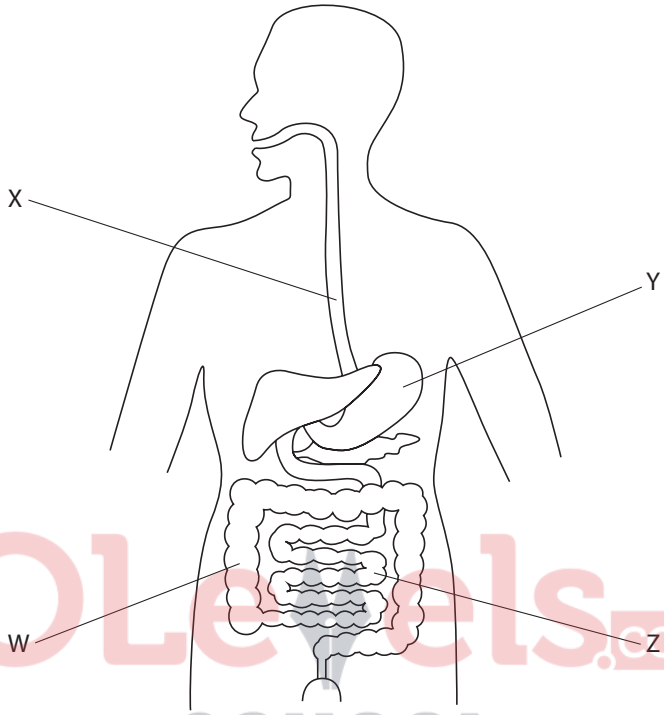
Working and answer

3. Question 2 · part a.ii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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P 7 0 9 4 9 A 0 6 2 4

3. Question 2 · part a.ii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

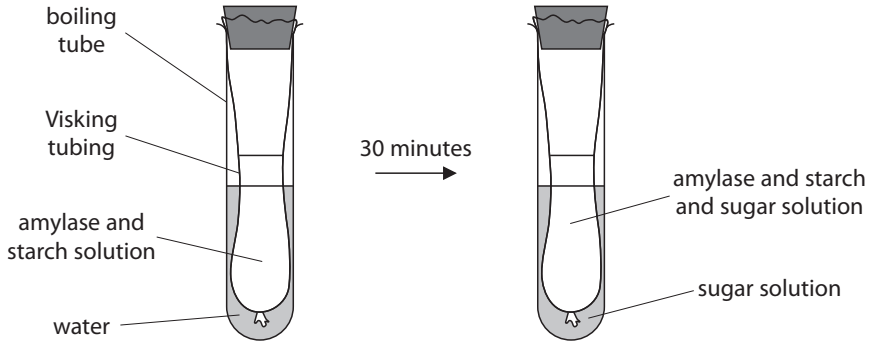
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3. Question 2 · part a.ii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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3. Question 2 · part a.ii · 4 marks · Working and answer

Working and answer

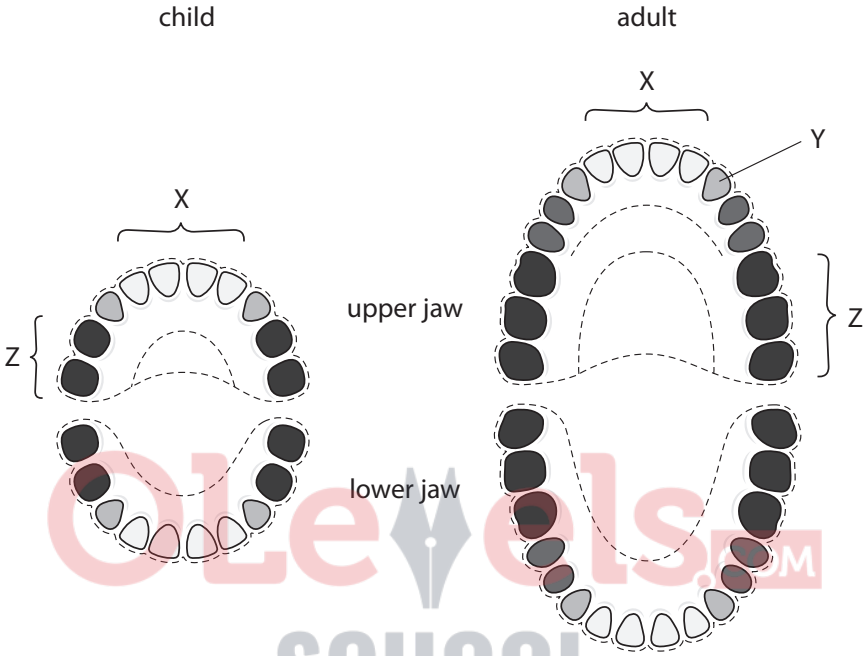
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4. Question 1 · part c.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

1 (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.



(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

(2)

X .....

Z .....

(ii) State the function of the tooth labelled Y.

(1)

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4. Question 1 · part c.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

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(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

(4)

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

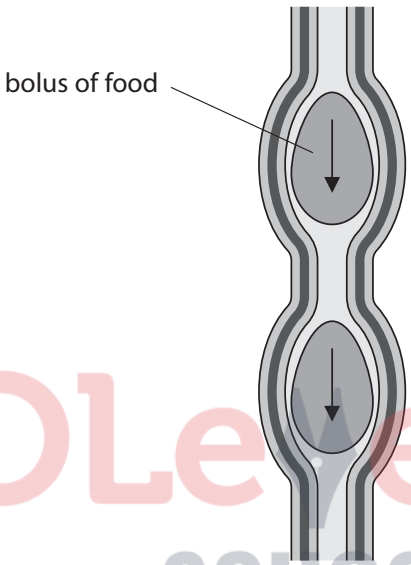
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4. Question 1 · part c.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

(c) The diagram shows the part of the digestive system that transports food from the mouth to the stomach.



(Source: https://www.motifolio.com/1031265.html)

(i) Name the part of the digestive system that transports food from the mouth to the stomach.

(1)

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(ii) Describe how the action of this part of the digestive system transports food from the mouth to the stomach.

(2)

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(Total for Question 1 = 10 marks)

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4. Question 1 · part c.ii · 2 marks · Working and answer

Working and answer

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